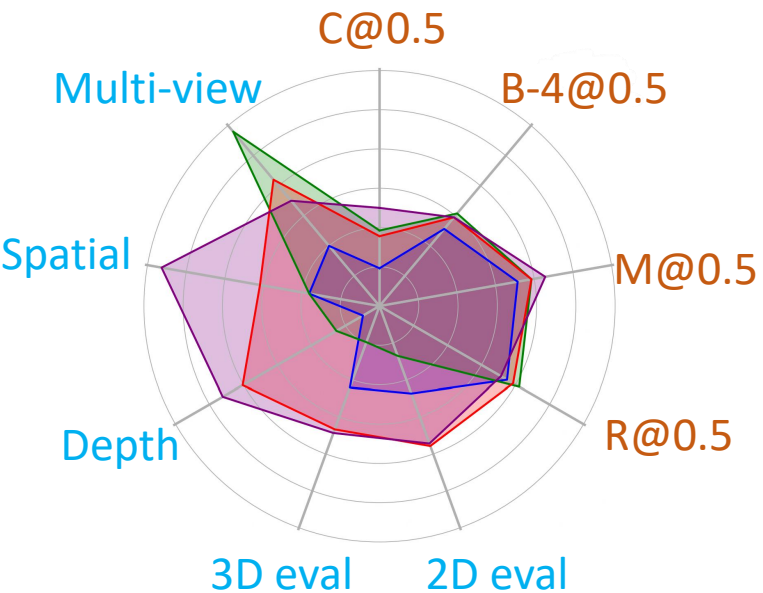


Randomzip Visionzip-3D Visionzip-2D Merge 3D

Spatial reasoning
bench

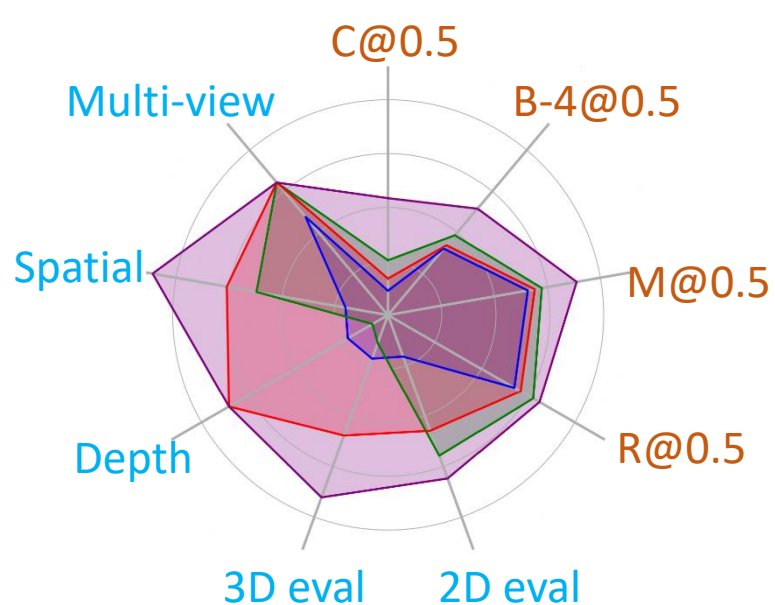
3D grounding
bench

Retain 30%



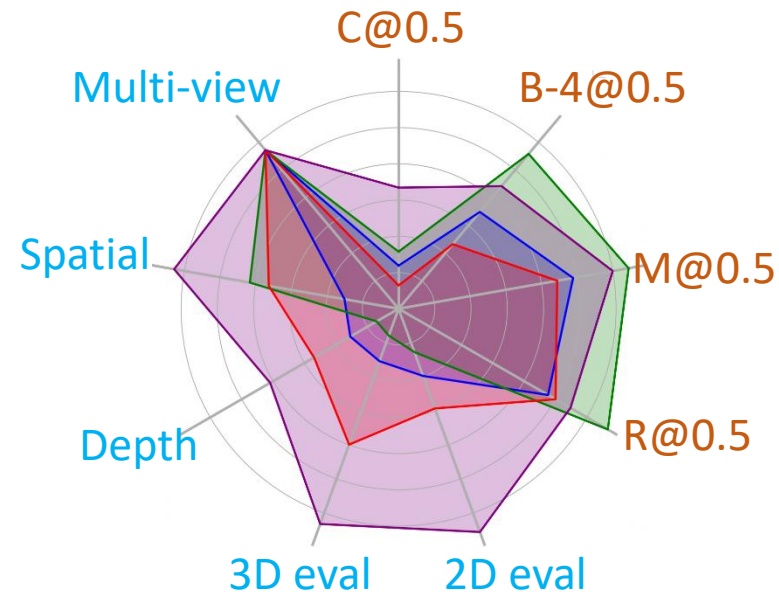
Merge3D: 97.5%
Visionzip-2D: 96.5%
Visionzip-3D: 94.0%
Randomzip: 92.9%

Retain 10%



Merge3D: 93.6%
Visionzip-2D: 90.7%
Visionzip-3D: 87.6%
Randomzip: 86.0%

Retain 5%



Merge3D: 89.3%
Visionzip-2D: 84.6%
Visionzip-3D: 84.2%
Randomzip: 83.3%